



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2025-26

Course Handout Part II

Date: 3rd August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : MEL G623
Course Title : ADVANCED VLSI DESIGN
Instructor-in-charge : PRASHANT KUMAR

- 1. Course description:** The field of Digital Very Large-Scale Integrated Circuits has gone through dramatic evolutions and changes. This course builds on the previous course MEL G621 VLSI DESIGN. It is intended to give a detailed knowledge and experience in the design of advanced VLSI circuits and chips in today's and future nano-scale CMOS technologies. Design of high-performance digital systems, failure issues, and remedies of problems. Topics will focus on current issues, including: wiring resistance and how to deal with it, power and Ground noise and regulation, clock (or asynchronous) system design and how to minimize clocking overhead, high-speed I/O design, energy minimization, including leakage control. Major VLSI design challenges are followed by careful treatment of several versatile digital and mixed analog-digital circuit building blocks frequently utilized in VLSI chips.
- 2. Scope and Objective:** The development of digital technology over the past 50 years has revolutionized the design and performance of everything from mobile phones to cars and entertainment systems; and engineers continue to enhance the performance of these devices. Recent developments in nanotechnology promise further large increases in processing power with lower power requirements, and will open up a whole new world of applications for digital techniques. The course aims to learn new techniques to keep up-to-date with developments in an industrial and/or research setting, and will have hands-on experience of the different stages of the design of a modern digital system.
- 3. Course learning outcomes:** Students will develop an understanding of common design and timing issues/challenges in digital systems. They will be introduced to high-speed/low-power circuit logic/ power minimization techniques, as well as wiring models used in digital systems. Additionally, they will learn about the clock system design and various noise sources that can interfere with these systems.

4 Text books and references:

TB: J. M. Rabaey, A. Chandrakasan, “Digital integrated Circuits-A design perspective”, Second Edition, Prentice Hall Electronics and VLSI Series.

RB1: IEEE journals

RB2: William J Dally, John W Poulton, " Digital Systems Engineering"; Cambridge University Press

RB3:A. Bellaour, M. Elmasry, “Low power digital VLSI design-circuits and systems”, Second Edition, Kluwer academic publishers,

RB4: R. Best, “Phase Locked Loop”, McGraw Hill Publishers.

RB5: S. S. Rofail, K. S. Yeo, “Low voltage, Low power Digital BiCMOS Circuits”, Prentice Hall Inc.

RB6: K. Roy, S.C. Prasad, “Low Power CMOS VLSI Circuit Design”, Wiley Interscience Publication.

RB7: M. J. S. Smith, “Application Specific Integrated Circuits”, Pearson Education (Singapore) Pte Ltd.

RB8: Bhaskhar Jayram, "AVHDL PRIMER", Prentice Hall.

RB9: IEEE Journals of solid-state circuits, VLSI system.

RB10: Miron Abramovici, Melvin A. Breuer, Arthur D. Friedman, “**Digital** Systems testing and testable design,” September 1994, Wiley-IEEE Press

RB11: Martin. Ken, “Digital Integrated Circuit Design”, Oxford University Press, Inc.

RB12: Michael. L. Bushnell, and Vishwani. D. Agrawal, “Essentials Of Electronic Testing For Digital, Memory and Mixed Signal VLSI Circuits. Kluwer Academic Publishers, Third Edition, 2004

RB13: Behrooz Parhami, “Computer Arithmetic-algorithms and Hardware design”, Oxford University Press Inc, 2000

RB14: Sutherland. I., Harris. David, Sproull. Bob, “Logical Effort-Designing Fast CMOS Circuits”, Morgan Kaufmann Publishers.

RB15: Eric Bogatin, " Signal integrity Simplified", Prentice Hall Modern Semiconductor Design Series

5 Method of conduct of the course: Through offline teaching (Blackboard/PPT slides)

6 Course Plan

Modules	Lecture No.	References	Learning Outcomes
General Introduction- digital system, data rate, pin count, serial/ parallel communication, eye diagram	1-2		Student will have Knowledge of System design issues that can cause non-functional system
Signal encoding, and signaling modes, voltage/ current Input/output drivers (with low power constraint) for C, and LRC loads	3-7	RB2, IEEE journals	Knowledge of signal encoding techniques, driver design, noise immunity



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Timing issues in synchronous VLSI systems.	8-15	TB (ch. 7) / IEEE journals	Knowledge of timing conventions and synchronization in digital systems
Asynchronous system design		TB (ch. 10) / IEEE journals	Knowledge of handshaking protocols in design
Wire design models in the nanometer region- lumped, distributed, transmission line, termination	16-22	RB2, TB (chapter 3), IEEE papers	Knowledge of Driver circuits design techniques, lumped/ transmission line modelling, signal integrity issues etc.
High-performance / low-power circuit logic, delay/ power minimization using logical effort	23-29	TB (chapter-14), IEEE papers	Knowledge of Techniques for high-speed circuits
Clock signals distribution and clock generation, clock drivers, Digital phase locked loop, digital delay locked loop, synchronizer - design and meta-stability issues	30-37	TB (Chapter - 10), RB3	Knowledge of clock distribution networks Generation of a clock using a PLL and its design
Noise management, power supply distribution	38-42	RB2, IEEE papers	Knowledge of noise sources-power supply, Crosstalk, inter-symbol interference, noise management techniques, Power distribution, regulation, isolation
Memory design (self-reading)		TB (Chapter 12), IEEE papers	Knowledge of Memory organization and design of memory cells

7. Evaluation Scheme

Component	Duration	Marks (100)	Date & Time	Evaluation type
Midsem Test	90 min	20 (20%)	05/10 FN2	Open Book
Research Project/ seminar / Viva - voce	(Continuous)	20 (20%)	Spread across the semester	Closed & Open Book
*Quiz	15 min	10 (10%)	To be announced	Open Book
**Assignments	-	20 (20%)	To be announced	Open Book
Comp. Exam	180 min	30 (30%)	11/12 AN	Closed Book

*Two quizzes, each carrying 5 marks, will be conducted. The combined score of the two quizzes will be considered for the final evaluation.

**Two take-home assignments, each carrying 10 marks, will be assigned.

Note: The Mid-Semester Examination will be conducted for 40 marks and will be scaled down to 20 marks for final grading. Similarly, the Comprehensive Examination will be conducted for 60 marks and will be scaled down to 30 marks final grading.

8. Chamber consultation hours: Saturday 11:00 AM – 12:00 PM

9. **Makeup Policy:** Make up will be given only on genuine reasons. Applications for makeup should be given in advance and prior permission should be obtained for Scheduled tests.

10. **Notices:** All notices related to the course will be put on NALANDA.

Instructor-in-charge
EEE (MEL-G623)